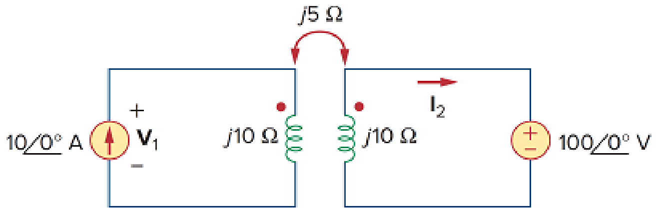
Chapter 13, Problem 6P

Problem

Given the circuit shown in Fig. 13.75, determine the value of V1 and I2.

Figure 13.75



Step-by-step solution

Step 1 of 4

Refer to Figure 13.75 in the textbook.

Consider the circuit shown in Figure 1.

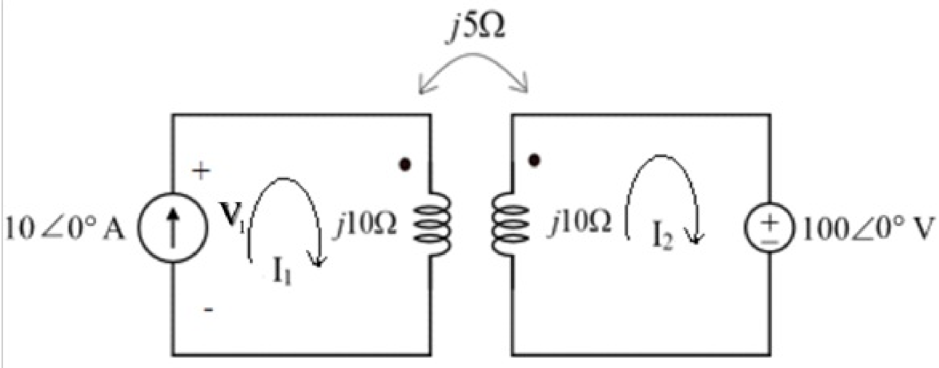


Figure 1

Step 2 of 4

Apply Kirchhoff's voltage law in loop 1.

$V_1 = j10I_1 - j5I_2$ ...... (1)

Apply Kirchhoff's voltage law in loop 2.

$$-j5I_1 + j10I_2 + 100\angle 0^{0} = 0$$
$$j10I_2 = j5I_1 - 100$$
$$I_2 = \frac{j5I_1 - 100}{j10}$$

$I_2 = 0.5I_1 + j10$ ...... (2)

Step 3 of 4

Determine the value of current I_2.

Substitute $10\angle 0^0$ A for I_1 in equation (2).

$$\begin{aligned} I_2 &= 0.5I_1 + j10 \\ &= 0.5\left(10\angle 0^0\right) + j10 \\ &= 5 + j10 \\ &= \sqrt{(5)^2 + (10)^2}\ \tan^{-1}\left(\frac{10}{5}\right) \end{aligned}$$

$$I_2 = 11.18\angle 63.43°\ \text{A}$$

Therefore, the value of current I_2 is $\boxed{11.18\angle 63.43°\ \text{A}}$.

Step 4 of 4

Determine the voltage V_1.

Substitute $(5 + j10)\ \text{A}$ for I_2, and $10\angle 0^0$ A for I_1 in equation (1).

$$\begin{aligned} V_1 &= j10I_1 - j5I_2 \\ &= j10(10\angle 0°) - j5(5 + j10) \\ &= j100 - j25 - (-50) \\ &= j75 + 50 \end{aligned}$$

$$\begin{aligned} V_1 &= 50 + j75 \\ &= 90.13\angle \tan^{-1}(1.5) \\ &= 90.13\angle 56.30°\ \text{V} \end{aligned}$$

Therefore, the voltage V_1 is $\boxed{90.13\angle 56.30°\ \text{V}}$.